Applying Explainability Techniques to Alzheimer's MRI Classification Models

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Introduction

Currently, over 55 million people are estimated to be living with Alzheimer's or other dementias, and this number is projected to be 139 million by 2050. The current standard methods of detection typically involve clinicians measuring different volumes of specific brain regions from MRI slices. Doing so is often time consuming, and potentially ignores greater complexities present in the entire MRI image.

As the capabilities of machine learning continue to grow and exceed expectations, particularly architectures involving deep neural networks, a large body of research has been dedicated to the detection of Alzheimer's disease (AD) from a wide variety of data sources using machine learning models. For example, Chui et al. were able to implement a custom GAN-CNN-TL architecture for MRI image classification capable of achieving 97.5% accuracy when predicting Alzheimer's disease in their test dataset (2022). Acharya et al. survey a total of 74 papers implementing a variety of deep learning architectures (CNN, ResNet, LSTM, Attention-based) using EEG signals (2025). Ebrahimi et al., were able to "inflate" the pretrained ResNet-18 for 2D image classification into a model capable of 3D image classification, and after some retraining with an MRI dataset, were able to achieve 96.88% accuracy (2020). The OASIS-3 (Open Access Series of Imaging Studies-3) dataset provides a longitudinal dataset of samples from a total of 4 classes (Normal, Very Mild AD, Mild AD, Moderate AD), integrating structural MRI, PET-based tauopathy (Braak staging), and longitudinal clinical assessments such as the Clinical Dementia Rating (CDR) and Montreal Cognitive Assessment (MoCA) (LaMontagne, 2019). From this dataset, we were able to obtain 1933 usable T1w MRI images with labels and additional clinical features.

While significant effort has gone towards creating these deep neural network classification models, Acharya et al. points out that much less effort has been done to increase the interpretability of their results (2025). Techniques for increasing explainability, such as Grad-CAM, have already been applied to CNN models built from medical imaging. Suara et al., for example, applied it to models built for detecting tumor patterns in the hopes of providing clinicians valuable insight into the decision making process of their model, ultimately increasing interpretability and therefore creating greater trust (2023). In this project, we seek to apply the same explainability techniques to the "inflated" 3D ResNet-18 model implemented by Ebrahimi et al., except we will retrain and evaluate using the larger, more nuanced OASIS-18 dataset, in hopes of gaining similar insight.

Methods

Data Preparation and Preprocessing

OASIS-3 is a longitudinal dataset which collected both imaging and clinical data in an effort to analyze the differences in the brain's changes over time from normal aging when compared to Alzheimer's disease. Using the following conversion – the same used by Chui et al. as is standard for Clinical Dementia Rating (CDR) scores – we were able to place each of our samples into one of 4 different classes: 0 -> Normal, 0.5 -> Very Mild, 1.0 -> Mild, 2.0+ -> Moderate. For each sample, we collected the following features of clinical data:

Clinical Data Features (OASIS-3)	
Name	**Description**
APOE (one-hot encoded)	23,24,33,34,44. Each is a genotype of the APOE gene, 23 being the genotype which shows some protection in developing Alzheimer's and 44 being the highest risk.
EDUC	Educational attainment, which is a score based on the highest level/years of schooling an individual has completed
momdem	0/1 – whether or not the mother of the participant had dementia
dadem	0/1 – whether or not the mother of the participant had dementia
AgeatEntry	Age when the participant enrolled in OASIS-18
Norm_Hippo	Normalized size of the hippocampus as measured from the MRI. Calculated by adding 'Left-Hippocampus_volume' and 'Right-Hippocampus_volume' and dividing by 'IntraCranialVol'
Norm_Entorhinal	Normalized size of the Entorhinal Cortex as measured from the MRI. Calculated by adding 'lh_entorhinal_volume' and 'rh_entorhinal_volume' and dividing by 'IntraCranialVol'
Sex (one-hot encoded)	Participant's sex

For the continuous features ('AgeatEntry', 'EDUC', 'Norm_Hippo', 'Norm_Entorhinal'), we applied MinMaxScaling. After dropping rows with missing columns, we have a total of 1933 samples from the 1098 participants in the study. Note that we only used data from the 2018 release as this was the same dataset used by Chui et al., whose methods we were originally attempting to replicate.

Our T1w MRI images (stored as NIfTI .nii.gz files) came in a variety of dimensions in the dataset due to the differences in equipment and procedures used throughout the OASIS-3 study. Therefore, we used a resampler, as provided in the SimpleITK Python library using a Linear Interpolator, in order to achieve a standard 224x224x224 image size for each scan while retaining the physical sizes (Beare, 2018).

Clinical Random Forest (Baseline)

We first implement a simple Random Forest Classifier using the clinical data features of our dataset. This establishes a baseline for classification accuracy to hopefully support our hypothesis that greater complexities garnered by deep learning models from the entire MRI image can lead to stronger, more accurate predictions than current practices. We split our dataset into 80% training and 20% testing subsets, applying stratification to ensure that the same ratio of different classes is seen in both datasets, which is important for proper evaluation given our imbalanced dataset (Normal: 1544, Very Mild: 311, Mild: 101, Moderate: 7). Given that we have a simple table of numerical discrete and continuous features, we were also able to apply SMOTE (Synthetic Minority Over-samping Technique) to generate synthetic samples of our minority classes in order to balance our dataset (Chawla et al., 2002). We specifically applied it to the training dataset after splitting to prevent any potential "leakage" of information, using k_neighbors=2 in the imbalanced learn Python library's implementation (Lemaître et al., 2017).

We used the Random Forest Classifier implementation from the Sci-kit Learn Python library using the resampled dataset outputted from SMOTE, using n_estimators=100 with all other hyperparameters left at their defaults (Pedregosa et al., 2011). Once trained, we are able to evaluate and compute test accuracy, a Confusion Matrix, and access feature importances for greater insight.

Proposed Architecture: "Inflated" ResNet-18 for MRI

For our proposed deep learning architecture, we apply the same "inflated" 3D ResNet-18 used by Ebrahimi et al. with success on the smaller (just 264 subjects) Alzheimer's Disease Neuroimaging Initiative (ADNI) dataset (2020). For the architecture, the inflation is done simply by extending the dimensions of the functional layers, and then adjusting the dimensions of the surrounding layers to fit. Specifically, every 2D convolutional layer becomes a 3D convolutional layer, all pooling layers become 3D pooling layers, and 2D batch normalization layers become 3D batch normalization. Ebrahimi et al. provides the following diagram for visualization (2020):

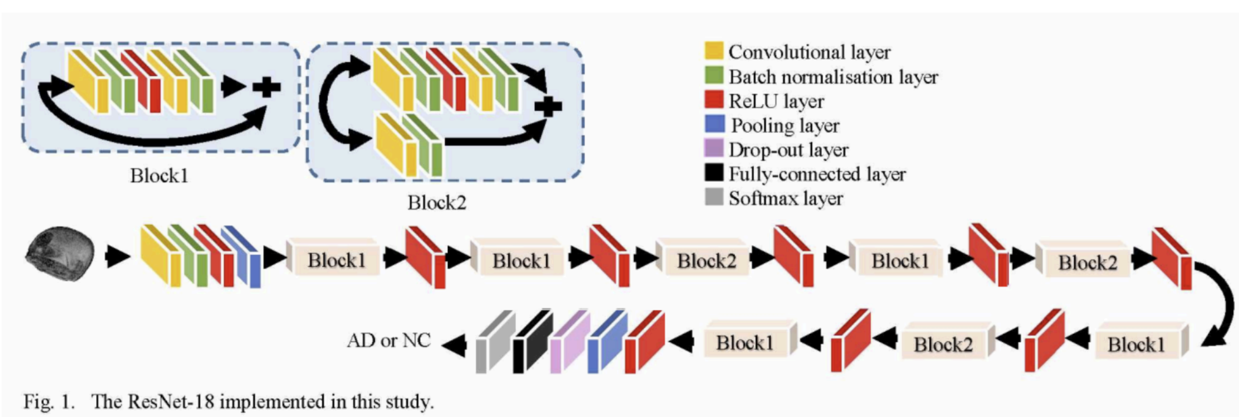

Figure 1: ***Architecture of the "Inflated" 3D ResNet-18.***

In this diagram, two types of residual "blocks" are illustrated, one containing an identity skip, and other one containing a projection skip. In the original ResNet paper, identity skips are proposed for when the dimensions of the input and output match such that $y = F(x) + x$ is defined. Projection skips, however, are proposed when there is a mismatch such that the input needs an operation with the minimum amount of additional learnable parameters (typically a 1 x 1 convolution, but in our case, a 1 x 1 x 1 convolution) to match dimensions with the output, $y = F(x) + P(x)$ (He et al., 2016). This is done specifically in Block2 because the first convolutional layer has an adjusted stride of 2, making the projection of the original input necessary. We provide an ordered list of all 71 layers, including dimensions, that we implemented using the PyTorch library (Paszke et al., 2026):

Stem

1. Conv3D (7×7×7, stride=2) → [N, 64, 112, 112, 112]
2. BatchNorm3D → [N, 64, 112, 112, 112]
3. ReLU → [N, 64, 112, 112, 112]
4. MaxPool3D (3×3×3, stride=2) → [N, 64, 56, 56, 56]

Layer1 (res2) - Block1

5. Conv3D (3×3×3) → [N, 64, 56, 56, 56]
6. BatchNorm3D → [N, 64, 56, 56, 56]
7. ReLU → [N, 64, 56, 56, 56]
8. Conv3D (3×3×3) → [N, 64, 56, 56, 56]
9. BatchNorm3D → [N, 64, 56, 56, 56]
10. **Add** → [N, 64, 56, 56, 56]
11. ReLU → [N, 64, 56, 56, 56]

Layer1 (res2) - 2nd Block1

12. Conv3D (3×3×3) → [N, 64, 56, 56, 56]
13. BatchNorm3D → [N, 64, 56, 56, 56]
14. ReLU → [N, 64, 56, 56, 56]
15. Conv3D (3×3×3) → [N, 64, 56, 56, 56]
16. BatchNorm3D → [N, 64, 56, 56, 56]

17. **Add** → [N, 64, 56, 56, 56]
18. ReLU → [N, 64, 56, 56, 56]

Layer2 (res3) - Block2

19. Conv3D (3×3×3, stride 2) → [N, 128, 28, 28, 28]
20. BatchNorm3D → [N, 128, 28, 28, 28]
21. ReLU → [N, 128, 28, 28, 28]
22. Conv3D (3×3×3) → [N, 128, 28, 28, 28]
23. BatchNorm3D → [N, 128, 28, 28, 28]
    Skip path
24. Conv3D (1×1×1, stride=2) → [N, 128, 28, 28, 28]
25. BatchNorm3D → [N, 128, 28, 28, 28]
26. **Add** → [N, 128, 28, 28, 28]
27. ReLU → [N, 128, 28, 28, 28]

Layer2 (res3) - Block1

28. Conv3D (3×3×3) → [N, 128, 28, 28, 28]
29. BatchNorm3D → [N, 128, 28, 28, 28]
30. ReLU → [N, 128, 28, 28, 28]
31. Conv3D (3×3×3) → [N, 128, 28, 28, 28]
32. BatchNorm3D → [N, 128, 28, 28, 28]
33. **Add** → [N, 128, 28, 28, 28]
34. ReLU → [N, 128, 28, 28, 28]

Layer3 (res4) - Block2

35. Conv3D (3×3×3, stride=2) → [N, 256, 14, 14, 14]
36. BatchNorm3D → [N, 256, 14, 14, 14]
37. ReLU → [N, 256, 14, 14, 14]
38. Conv3D (3×3×3) → [N, 256, 14, 14, 14]
39. BatchNorm3D → [N, 256, 14, 14, 14]
    Skip path
40. Conv3D (1×1×1, stride=2) → [N, 256, 14, 14, 14]
41. BatchNorm3D → [N, 256, 14, 14, 14]
42. **Add** → [N, 256, 14, 14, 14]
43. ReLU → [N, 256, 14, 14, 14]

Layer3 (res4) - Block1

44. Conv3D (3×3×3) → [N, 256, 14, 14, 14]

45. BatchNorm3D → [N, 256, 14, 14, 14]
46. ReLU → [N, 256, 14, 14, 14]
47. Conv3D (3×3×3) → [N, 256, 14, 14, 14]
48. BatchNorm3D → [N, 256, 14, 14, 14]
49. **Add** → [N, 256, 14, 14, 14]
50. ReLU → [N, 256, 14, 14, 14]

Layer4 (res5) - Block2

51. Conv3D (3×3×3, stride=2) → [N, 512, 7, 7, 7]
52. BatchNorm3D → [N, 512, 7, 7, 7]
53. ReLU → [N, 512, 7, 7, 7]
54. Conv3D (3×3×3) → [N, 512, 7, 7, 7]
55. BatchNorm3D → [N, 512, 7, 7, 7]

    Skip path

56. Conv3D (1×1×1, stride=2) → [N, 512, 7, 7, 7]
57. BatchNorm3D → [N, 512, 7, 7, 7]
58. **Add** → [N, 512, 7, 7, 7]
59. ReLU → [N, 512, 7, 7, 7]

Layer4 (res5) - Block1

60. Conv3D (3×3×3) → [N, 512, 7, 7, 7]
61. BatchNorm3D → [N, 512, 7, 7, 7]
62. ReLU → [N, 512, 7, 7, 7]
63. Conv3D (3×3×3) → [N, 512, 7, 7, 7]
64. BatchNorm3D → [N, 512, 7, 7, 7]
65. **Add** → [N, 512, 7, 7, 7]
66. ReLU → [N, 512, 7, 7, 7]

Head

67. GlobalAvgPool3D → [N, 512, 1, 1, 1]
68. Flatten → [N, 512]
69. Fully Connected → [N, num_classes]
70. Softmax → [N, num_classes]
71. Classification output

Note that as an implementation detail, our first "layer" is actually 3D ZScoreNormalization such that different ranges of intensity values per MRI can be standardized more uniformly, as recommended by Carré et al. (2020).

In order to apply the pretrained parameters from the pretrained 2D ResNet-18 model to our 3D architecture, we first need to average the weights from the first convolutional layers' 3 RGB channels as our MRI images are just a single dimension. For convolutional layers, we repeat the weights across the new depth dimension but then divide by the size of the depth in order to keep the same level of activation for when all are summed for the next layers. For the BatchNorm layers, we can simply copy the weight, bias, running_mean, and running_var parameters directly.

For continued training using our preprocessed MRI images and labels from OASIS-3, we once again split our datasets into 80% training and 20% testing, once again applying stratification to ensure both subsets contain the same balance of classes. Furthermore, we also apply stratification to the sampling of our batches in the training set in order to reduce variance between our batches, improving our training's stability. Due to computational and time constraints, we were not able to apply a true method to tune our hyperparameters such as cross-validation, so we instead simply employed hyperparameters similar to Ebrahimi et al.'s: (Adam Optimizer, LR: 0.0001, Batch Size: 8, 40 Epochs).

Applying Explainability: GradCAM

GradCAM was applied by using the M3d-CAM Python Library, which allows injection into any PyTorch model and is capable of both producing 3D heatmaps and saving them in NIfTI format for easy use as overlays in a standard medical image viewer such as Mango (Gotkowski et al., 2020). We specifically chose to apply GradCAM to the second convolutional layer in

Layer3's Block1 (number 47 in our ordered list), as we found this to be a good balance between capturing the deeper-level complex features of the model and not needing to upscale the map so much that the heatmap would be blurry. GradCAM heatmaps were ultimately computed when evaluating using the test dataset using the gradient of the predicted class score with respect to the feature maps of layer 47.

Results

Comparative Performance Metrics: Accuracy and Sensitivity

The 3D ResNet-18 model achieved a final test accuracy of 78.29%. To evaluate this performance, we benchmarked the deep learning model against a Random Forest classifier trained on clinical volume data (e.g., hippocampal and entorhinal volume). The Random Forest model achieved a similar accuracy of 79.39%.

This comparison is significant because it demonstrates that our 3D ResNet-18 is performing at a level comparable to traditional clinical methods. While the Random Forest relies on pre-measured features/volumes, the ResNet-18 is performing an end-to-end feature extraction directly from raw 3D pixels.

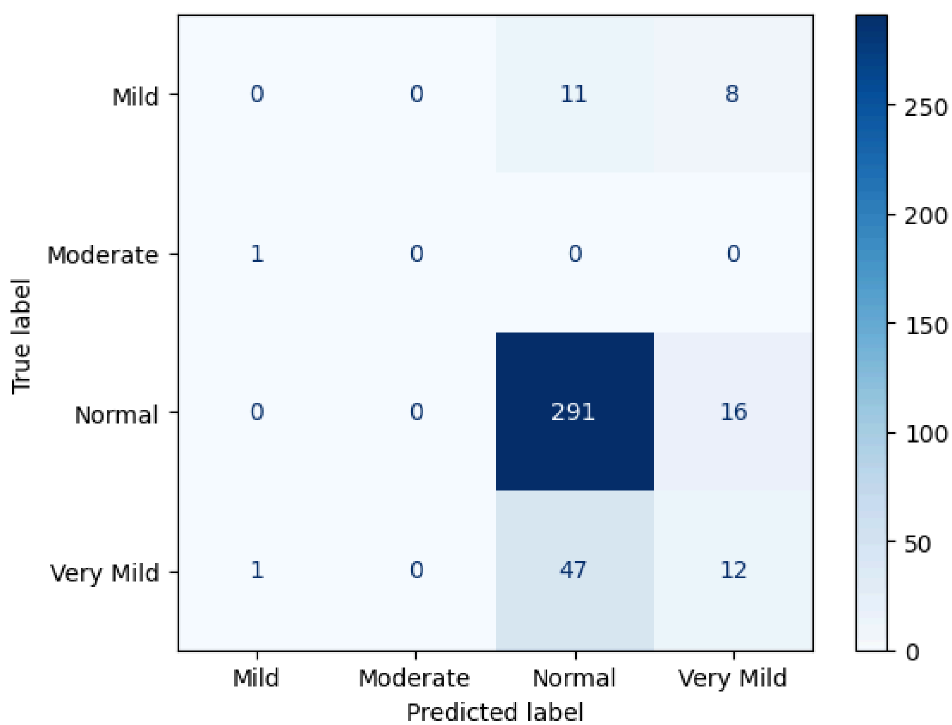

Figure 2: ***Confusion Matrix***

The Confusion Matrix provides a deeper look at class sensitivity. The model was excellent at identifying "Normal" cases, correctly identifying 291 out of 307 healthy subjects. However, the results show significant confusion between the Normal and Very Mild classes. This suggests that the model struggles to distinguish the subtle, overlapping morphological changes that occur in the earliest stages of cognitive decline. This performance was heavily influenced by the dataset imbalance, where healthy scans outnumbered moderate cases.

Training Dynamics and Convergence Volatility

The training logs for the 3D ResNet-18 revealed a downward trend in the Loss Curve, indicating that there was successful optimization. However, the Accuracy Curve was seen to have significant "jitter" or high volatility throughout the training epochs.

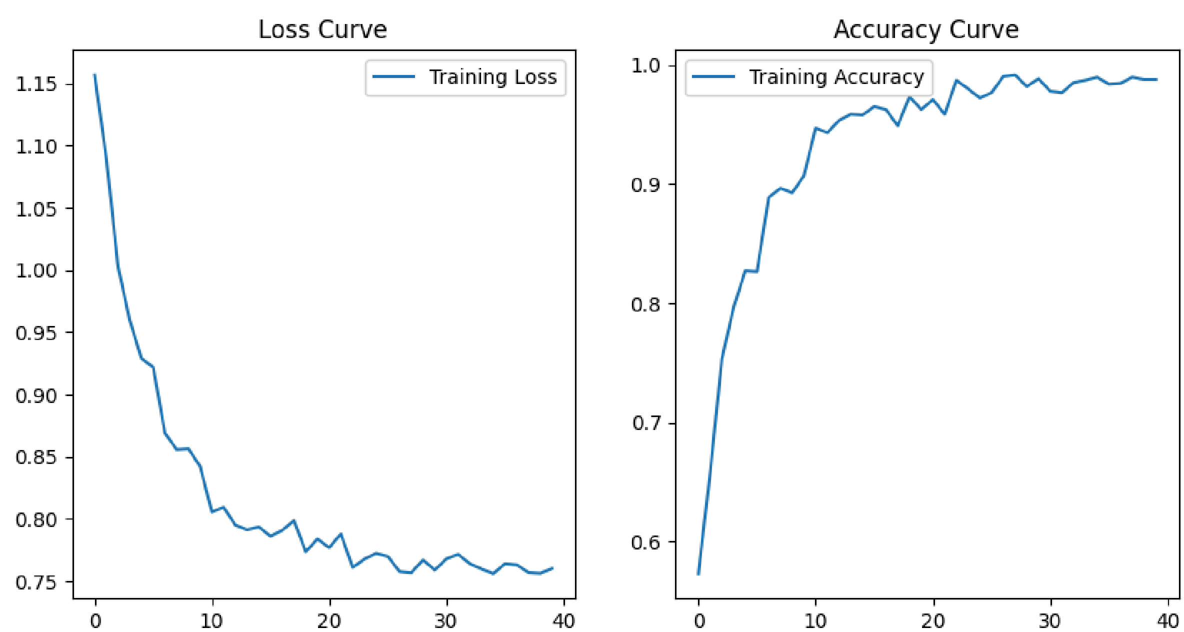

Figure 3: ***Loss and Accuracy Curves.***

In an ideal deep learning training scenario, the validation accuracy curve should follow a smooth trajectory, demonstrating a rapid initial increase followed by a steady flattening or plateau as the model converges on an optimal solution. However, our results showed persistent volatility rather than a smooth stabilization.

In 3D deep learning, such volatility often indicates Noise Sensitivity. Because the model was processing the entire 3D volume, including the skull, eyes and scalp, the high variance between subjects' facial structures created "noise" that the model struggled to generalize. This volatility indicated that the model was likely latching onto extracranial features rather than stable neurological markers.

Diagnostic Auditing: Identification of Ocular Bias.

To investigate the underlying causes of the training volatility, Grad-Cam was implemented to visualize the model's spatial localization and attention heatmaps. Our audit revealed a consistent Ocular Bias (Figure 4). In a high percentage of cases, the model's highest activation regions were located over the eyes and facial structures rather than the internal brain tissue.

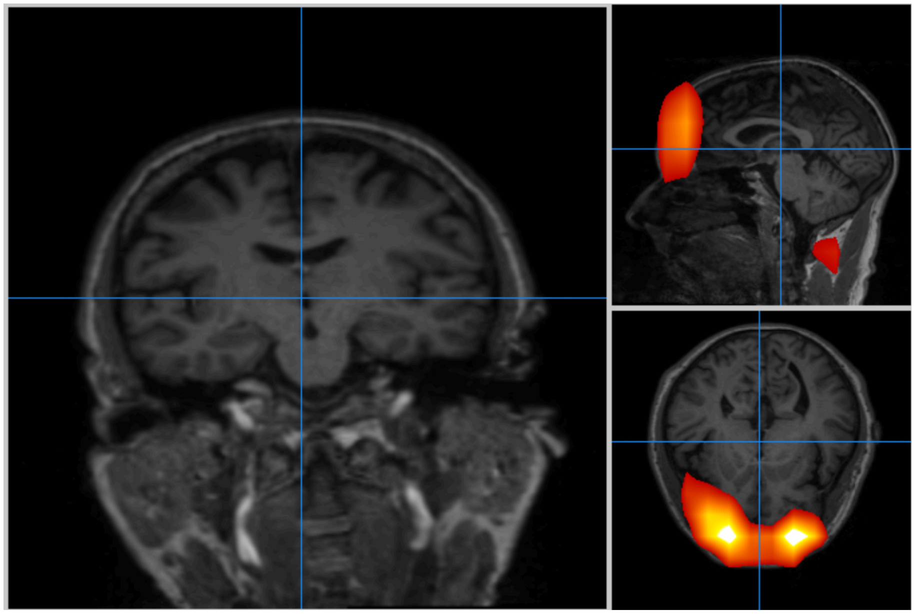

Figure 4: ***Grad-CAM visualizations in MANGO.*** *Despite correct classification ("Healthy"), the high-activation regions are centered on extracranial facial features.*

This "shortcut learning" explains both the model's success in healthy cases and its training jitter. Since facial structures are highly variable but distinct, the model was using them to differentiate subjects. This finding highlights the limitation that the model was right for the wrong reasons. This realization showed the necessity of 3D skull-stripping in future iterations to eliminate extracranial features and force the network to prioritize intracranial anatomy.

Proof of Concept: Moderate Case

Despite the ocular bias found in earlier stages, a diagnostic audit was conducted on a subject with Moderate AD to determine if the 3D ResNet-18 architecture could localize true pathological markers. In clinical neurology, the hippocampus (located in the Medial Temporal Lobe) is considered the "gold standard" biomarker for Alzheimer's Disease. According to the established model of the Alzheimer's pathological cascade (Jack et al.., 2010), hippocampal volume loss is one of the earliest and most significant indicators of neurodegenerative atrophy, often preceding the onset of clinical dementia. Our model's ability to localize attention to this specific region in the Moderate case (Figure 5) suggests that the RestNet-18 is capturing the same structural decline that clinicians use for manual diagnosis.

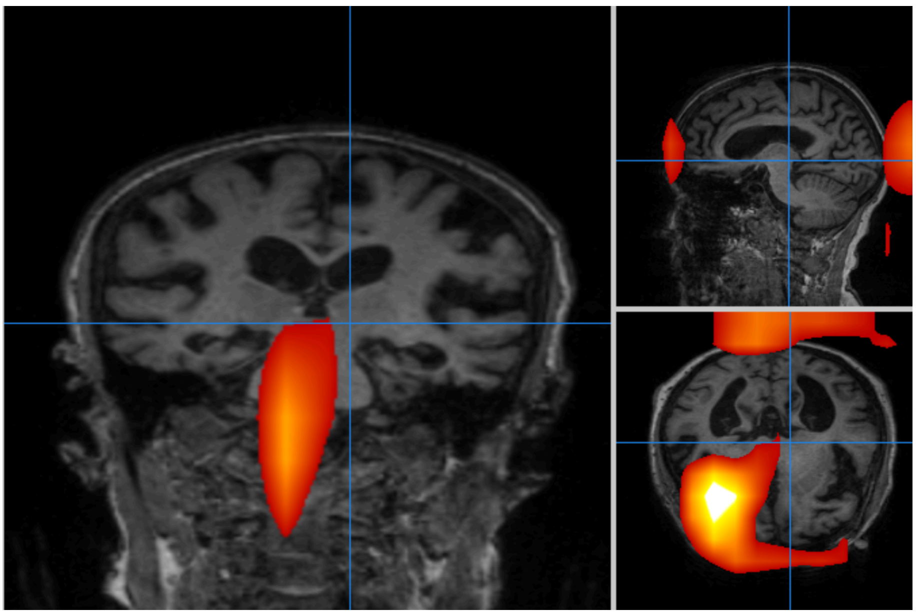

Figure 5: ***3D Grad-CAM overlay in Mango*** *showing localized attention near the Hippocampus in a Moderate Alzheimer's patient.*

As illustrated in Figure 5, the Grad-CAM activation for the Moderate case shifted significantly away from the facial structures and toward the internal cranium. The model's "attention" localized to regions in close proximity to the hippocampus and the surrounding temporal cortex. This localization was verified using MANGO (Multi-Image Analysis GUI) to ensure that the 3D heatmap coordinates correspond accurately with the medial temporal lobe.

This shift serves as a critical Proof of Concept. It demonstrates that the inflated 3D ResNet-18 architecture is capable of identifying valid neuroanatomical biomarkers rather than relying solely on facial structures.. This finding validates the model's potential as a diagnostic tool; however, it also ensures that an elimination of facial features through automated skull-stripping is an essential prerequisite for this hippocampal localization to become a consistent, reliable behavior across all stages of the disease.

Discussion

Interpretation of Results and "Shortcut Learning"

The most critical finding of this research is the identification of Ocular Bias, a form of "shortcut learning" where the 3D ResNet-18 prioritized extracranial facial features over internal neuroanatomy. As defined by Geirhos et al. (2020), shortcut learning occurs when a deep neural network identifies a consistent, easily accessible pattern in the training data to achieve high performance on a benchmark, rather than learning the intended biological features. This suggests that while the model was effective at classifying the OASIS-3 dataset, its decision-making process was fundamentally disconnected from the neurological reality of Alzheimer's Disease.

While our 3D ResNet-18 model achieved a competitive 78.29% accuracy, the Grad-CAM audit revealed that the model was essentially "cheating" by prioritizing the eyes and scalp over internal neuroanatomy. This highlights a critical vulnerability in medical AI, where high quantitative accuracy can mask a fundamental lack of clinical validity. Without the spatial explainability provided by Grad-CAM, the model's reliance on these non-biological features would have remained undetected.

Neuroanatomical Significance of the Hippocampus

To understand why our model's shift in attention was so important, we must look at the underlying pathology of Alzheimer's Disease. In clinical neurology, the hippocampus is considered the "gold standard" biomarker for AD. It is a small structure located in the Medial Temporal Lobe that is responsible for memory. According to the research by Jack et al. (2010), the hippocampus is usually one of the first places to shrink (atrophy) when someone has

Alzheimer's. This means that if an AI is actually working, that is exactly where it should be looking.

In our project, we compared two very different ways of looking at the brain. Our Random Forest model acted like a "spreadsheet of the brain." We fed pre-measured clinical volumes, and the Feature Importance graph confirmed that "Norm_Hippo" (normalized hippocampal volume) was the #1 most important feature for making a correct diagnosis. This gave us our clinical ground truth.

On the other hand, our 3D ResNet-18 relied on "Spatial Awareness." Instead of a "spreadsheet of the brain," it looks at raw 3D pixels to find patterns that a human might miss. In our Moderate AD proof-of-concept, the Grad-CAM heatmaps localized on the internal cranium, specifically targeting areas of the temporal lobe and hippocampal region.

When we put these two findings together, the results are compelling. The "spreadsheet" (Random Forest) and the "spatial awareness" (ResNet-18) both prioritized the Medial Temporal Lobe. This alignment proves that the 3D ResNet-18 is capturing the same structural decline that clinicians use for a manual diagnosis. It suggests that while the model may get distracted by the face in early stages, the architecture itself is fundamentally sound and capable of seeing the "clinical truth" once the pathology is strong enough.

Technical Limitations and Computational Constraints

One challenge we faced throughout the whole project was an imbalance of classes in our dataset. While we were able to implement SMOTE for our RandomForestClassifier baseline with some success, we lacked the additional training time to implement a deep learning equivalent capable of generating entire MRI images of our minority class. In their work, Chui et al. actually implement a Generative Adversarial Network (GAN) as part of their entire model's pipeline capable of doing just that (2022). However, because many of the details of their implementation (hyperparameters, layers, etc.) were left out, implementing it ourselves may not have been as straightforward as we initially thought, especially after we struggled achieving comparable results with just the base custom CNN architecture.

Yet another challenge we faced with the 3D ResNet-18 was the sheer size of the model and its inputs. Training with a batch size of 8, the model needed to allocate over 50 GB of VRAM in the GPU during training. Issues with garbage collection from the surrounding code also meant that we ran into several "insufficient memory" errors when retraining after adjusting hyperparameters. Furthermore, we needed to use one of the highest runtimes offered by Google Colab, the G4, which quickly burned the limited compute units we had available as each training took about 55 minutes. While we were able to do some tuning of hyperparameters from our limited trials, this limited us from performing a true method of Cross Validation in order to further tune into the optimal values for this particular OASIS-3 dataset.

Future Work: Roadmap to Clinical Readiness

To transition our research from a proof-of-concept into a reliable diagnostic tool, we have a developed three-pillared roadmap for future development.

1. **Implementing 3D Skull-Stripping**

   The most immediate priority is solving the "Ocular Bias" we discovered during our Grad-CAM audit. To do this, we need to implement a 3D Skull-Stripping factory. This is a preprocessing step that uses specialized algorithms to strip away all non-brain tissue from the MRI volumes before they reach the model. By cleaning the scans this way, we can effectively block the model from taking shortcuts. It forces the network to focus entirely on the internal anatomy of the brain, ensuring that the classification is based on actual neurology rather than facial structures.

2. **Balancing the Dataset with MONAI Generative AI**

   Currently it is difficult for our model to learn what "Moderate Alzheimer's" looks like because it barely has any examples to study. To put into perspective, our current data split is split like this:

   - Healthy (Normal): 1,544 scans
   - Very Mild: 311 scans
   - Mild: 101 scans
   - Moderate: Only 7 scans

   This means that for every one "Moderate" brain the model saw, it saw over 200 healthy brains. Because of this extreme gap, the model developed conservative bias, as it would often guess "Normal" because, statistically, it was the safest bet to keep the overall accuracy high.To solve this "Minority Class" problem, we plan to use a framework called MONAI (Medical Open Network for AI). MONAI is specifically designed for medical imaging and includes tools for Generative AI. We can use these tools to generate synthetic, anatomically accurate MRI scans of patients with "Moderate" and "Mild" Alzheimer's. By creating these synthetic scans, we can equal out our dataset. This gives the model enough examples of real pathology to learn from, making it more sensitive to the subtle changes in the brain that we currently struggle to detect.

3. **Moving to Continuous MoCA Score Regression**

   Finally, we want to change how the model actually thinks about Alzheimer's. Right now it is a multi-classification tool that predicts the disease . In the future, we want to move toward a 3D Regression task that predicts a patient's continuous MoCA (Montreal Cognitive Assessment) score.

   The MoCA is a standard 30-point test clinicians use to measure cognitive decline. By training the model to predict a specific score rather than just a category, we can give doctors more of a granular look at a patient's health. This would allow them to track the specific rate of a patient's decline over time.

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Usage of AI

Gemini: AI Search for quicker finding of relevant sources. Basic code/framework for scaffolding of data preparation, model initialization, and final evaluation within Google Colab

ChatGPT: Transcription of Ebrahimi et al.'s Matlab code which "inflates" both the architecture and pretraining weights of ResNet-18 for 3D CNN into Python using PyTorch